

Esperanza de una función de una variable aleatoria

Proposición 1. Sea $g: \mathbb{R} \rightarrow \mathbb{R} \in \mathcal{L}^1(\mu_X)$ y $X: \Omega \rightarrow \mathbb{R}$ una variable aleatoria

$$\implies \mathbb{E}[g(X)] = \int_{\mathbb{R}} g(t) d\mu_X(t).$$

Demostración: Dividiremos la demostración en pasos:

(1) Consideramos primero $g = \mathbb{1}_A$, con $A \in \mathcal{B}(\mathbb{R})$, entonces $g(X) = \mathbb{1}_A(X)$

$$\begin{aligned} \implies \mathbb{E}[g(X)] &= \int_{\Omega} \mathbb{1}_A(X(\omega)) d\mathbb{P}(\omega) = \int_{\Omega} \mathbb{1}_{\{X(\omega) \in A\}} d\mathbb{P}(\omega) \\ &= \mathbb{P}(X \in A) = \mu_X(A) = \int_{\mathbb{R}} \mathbb{1}_A(t) d\mu_X(t) = \int_{\mathbb{R}} g(t) d\mu_X(t). \end{aligned}$$

(2) Por tanto, si $g = \sum_j \alpha_j \mathbb{1}_{E_j}$ es simple, entonces por linealidad de la integral:

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_{\Omega} g(X(\omega)) d\mathbb{P}(\omega) = \int_{\Omega} \left(\sum_{j=1}^n \alpha_j \mathbb{1}_{E_j}(X(\omega)) \right) d\mathbb{P}(\omega) = \sum_{j=1}^n \alpha_j \int_{\Omega} \mathbb{1}_{E_j}(X(\omega)) d\mathbb{P}(\omega) \\ &\stackrel{??}{=} \sum_{j=1}^n \alpha_j \int_{\mathbb{R}} \mathbb{1}_{E_j}(t) d\mu_X(t) = \int_{\mathbb{R}} \left(\sum_{j=1}^n \alpha_j \mathbb{1}_{E_j}(t) \right) d\mu_X(t) = \int_{\mathbb{R}} g(t) d\mu_X(t). \end{aligned}$$

(3) Entonces, si g es medible no negativa, por un lema técnico, $\exists (s_n)_{n \in \mathbb{N}}$ una sucesión no decreciente de funciones simples tales que $\forall t \in \mathbb{R} : \lim s_n(t) = g(t)$, entonces por convergencia monótona:

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_{\Omega} g(X(\omega)) d\mathbb{P}(\omega) = \int_{\Omega} \lim_{n \rightarrow \infty} s_n(X(\omega)) d\mathbb{P}(\omega) \\ &\stackrel{\text{TCM}}{=} \lim_{n \rightarrow \infty} \int_{\Omega} s_n(X(\omega)) d\mathbb{P}(\omega) \stackrel{??}{=} \lim_{n \rightarrow \infty} \int_{\mathbb{R}} s_n(t) d\mu_X(t) \stackrel{\text{TCM}}{=} \int_{\mathbb{R}} g(t) d\mu_X(t). \end{aligned}$$

(4) Finalmente, si g es medible arbitraria, entonces $g = g^+ - g^-$, con g^+, g^- medibles no negativas, y por linealidad de la integral:

$$\int_{\mathbb{R}} g(t) d\mu_X(t) = \int_{\mathbb{R}} g^+(t) d\mu_X(t) - \int_{\mathbb{R}} g^-(t) d\mu_X(t) \stackrel{??}{=} \mathbb{E}[g^+(X)] - \mathbb{E}[g^-(X)] = \mathbb{E}[g(X)].$$

■

Referenciado en

- `obs-linealidad-esperanza`
- `cor-formula-esperanza`

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.